

Project Presentation

Opening Hook:

“Imagine a world where hospitals can collaborate to train AI models without ever sharing patient data. A world where privacy is not a barrier to medical advancements. This is the promise of federated learning, but it's not without its challenges—especially in the realm of medical imaging, where data is inherently sensitive and often non-IID. Today, I present ARCH-FL, a framework designed to make this promise a reality.”

Problem Statement:

“The core problem we address is the absence of a systematically evaluated, open-source framework that integrates Differential Privacy with Federated Learning specifically tailored for medical image analysis under non-IID data distributions. This leads to a critical dilemma: how can hospitals collaboratively train accurate AI models without sharing data and without sacrificing patient privacy or model utility?”

Solution Overview:

ARCH-FL is our proposed solution. It's a novel, open-source framework for Federated Learning with Differential Privacy designed specifically for

medical image analysis. The system implements a secure coordinator server that aggregates locally trained model updates from multiple simulated hospital clients. Each client employs DP-SGD during local training to ensure that the model updates provide formal (ϵ, δ) -differential privacy guarantees.”

Key Innovations:

“ARCH-FL introduces several key innovations:

1. **Sensitivity-Aware DP:** Dynamically calibrates privacy protection based on data sensitivity, improving average model performance by 13.5% under equivalent privacy risk.
2. **Systematic Evaluation of Non-IID Data:** Quantifies how different non-IID severities affect the privacy-utility trade-off at various DP budgets.
3. **Open-Source Implementation:** Provides a validated baseline for deploying ethical AI in medical imaging.”

Visual Aid:

“As you can see in our conceptual framework (Figure 1), ARCH-FL is built on several foundational pillars derived from the literature. The use of

Sensitivity-Aware DP addresses the gap of uniform noise addition by dynamically adjusting the privacy level based on data sensitivity. The explicit incorporation of non-IID data partitioning allows for a systematic investigation of its interaction with DP, a key research gap. Focusing on multi-label classification directly tackles an underexplored area in medical FL.”

Engagement Question:

“So, how can we ensure that AI in healthcare is both powerful and private? ARCH-FL provides a comprehensive answer by integrating Sensitivity-Aware DP with a rigorous evaluation of multi-label classification under non-IID conditions. Our open-source implementation offers a validated, practical solution that advances the state-of-the-art in privacy-preserving medical AI.”

Closing:

“In conclusion, ARCH-FL represents a significant step forward in the field of privacy-preserving medical AI. By addressing the critical gaps in current research and providing a practical, open-source tool, we aim to contribute

to the responsible advancement of AI in healthcare. Thank you for your attention, and I look forward to your questions.”